

Cosift Open Contribution Network — Flow Diagrams

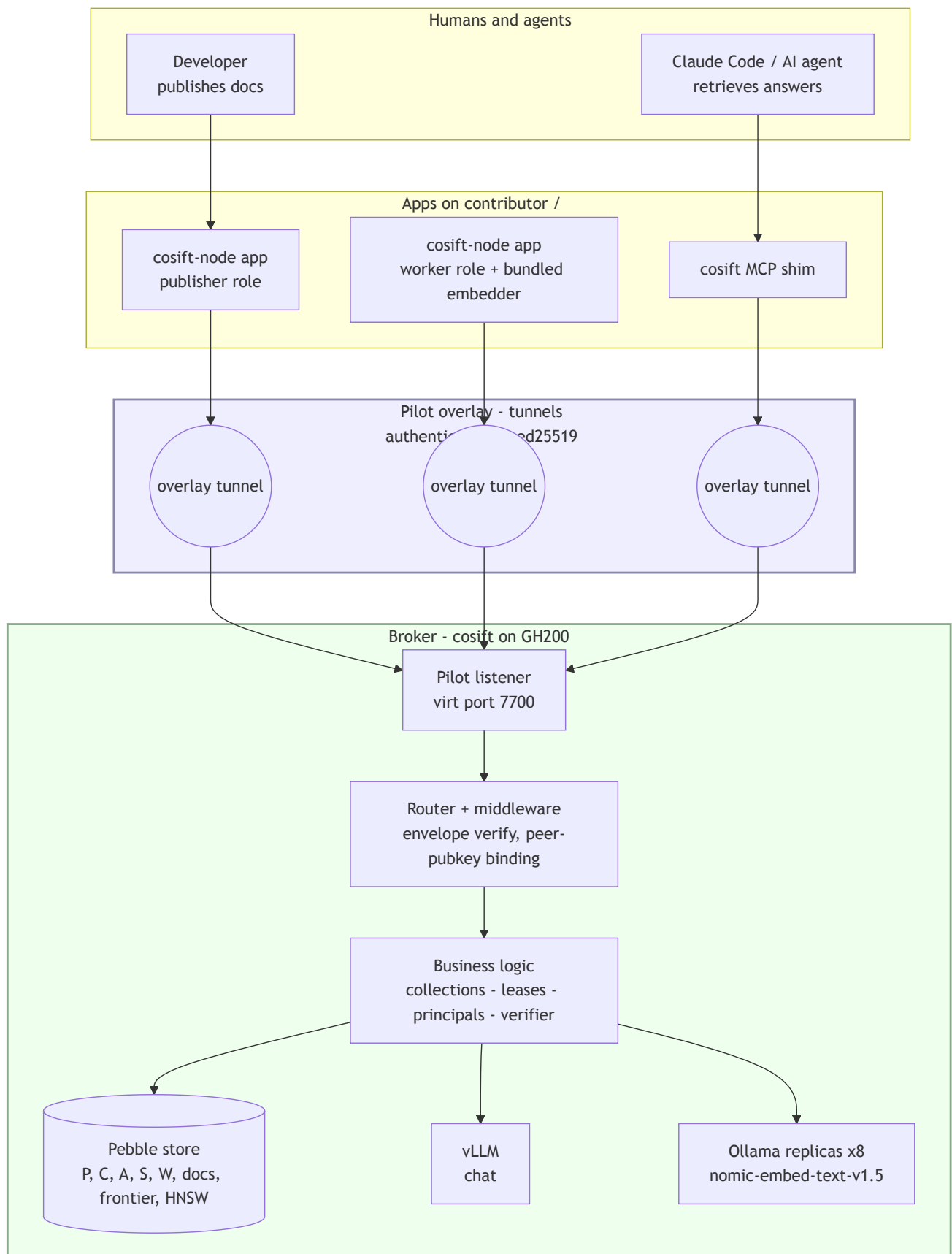
Companion to `PLAN-contrib-network.md`. Each diagram answers one question and references the workstreams (WS0-WS15) implementing it. Mermaid renders natively on GitHub.

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1. System overview

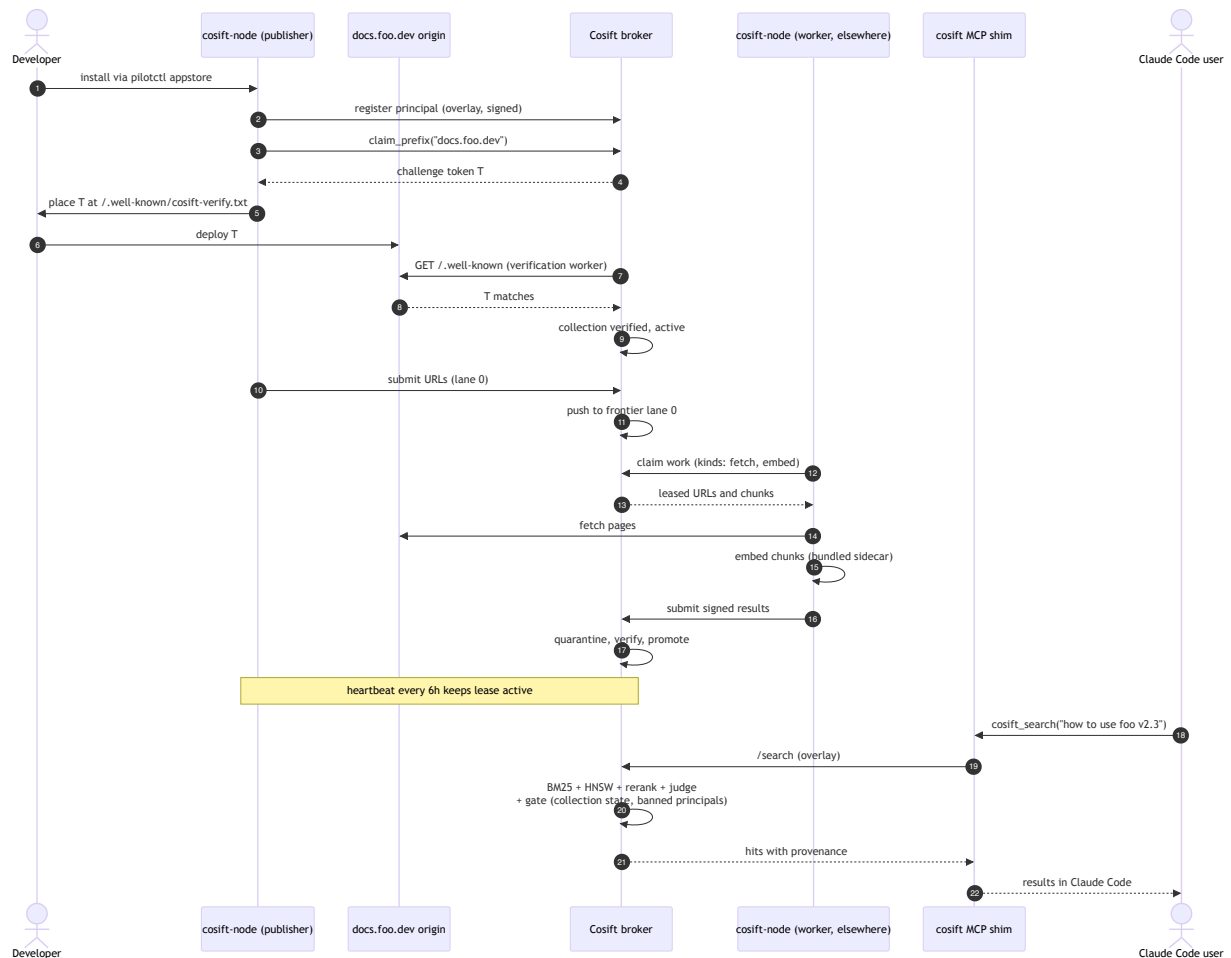
Three planes — humans/agents at the top, Pilot overlay in the middle, the broker (cosift on GH200) at the bottom. Every contribution byte crosses the middle plane (**I9**).



Trust boundaries: every arrow crossing into the broker is overlay-authenticated; the envelope signing key equals the tunnel peer pubkey.

2. End-to-end story

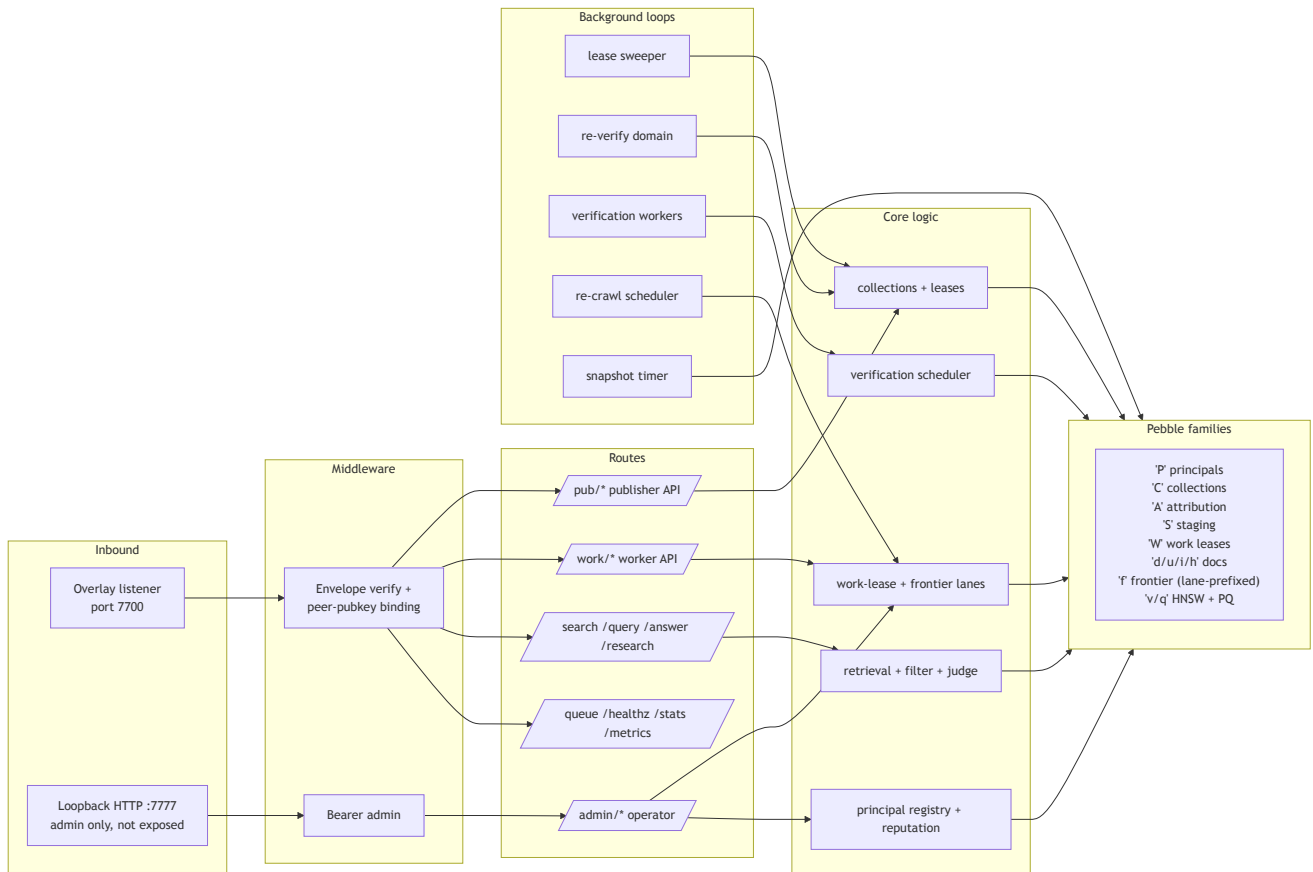
The full loop from a dev shipping docs to an AI agent retrieving an answer. Time flows top-to-bottom; only happy-path events shown.



The lease is what makes "active only while installed" true: the developer's app stops heartbeating when uninstalled, lease lapses, docs leave the index.

3. Component map of the broker

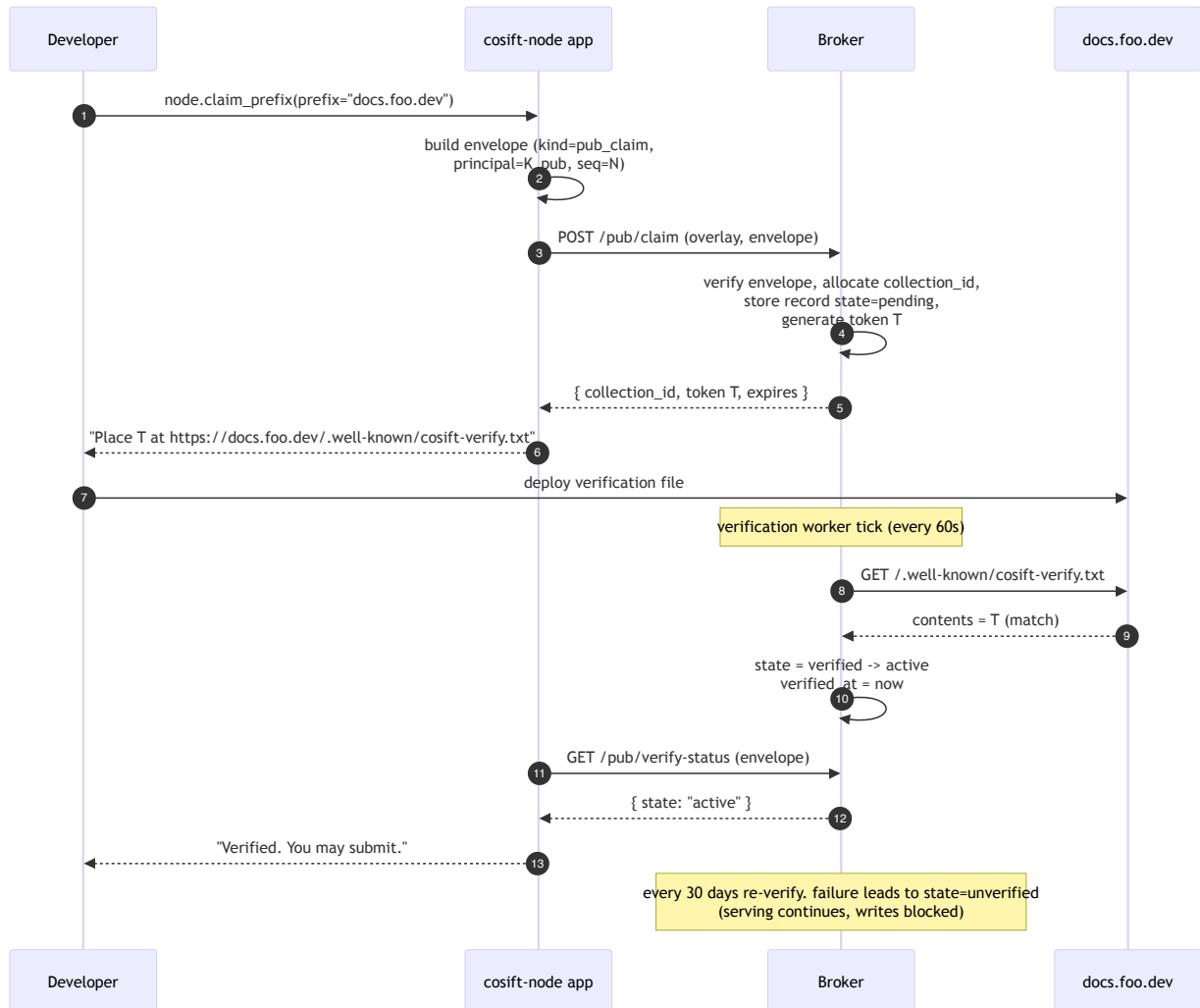
Zoom into one cosift instance. Routes split by auth surface; background loops sit on the side.



`/admin/*` routes only exist on loopback HTTP behind the operator's admin token — the overlay listener refuses them outright (route allowlist).

4. Publisher claim and verify

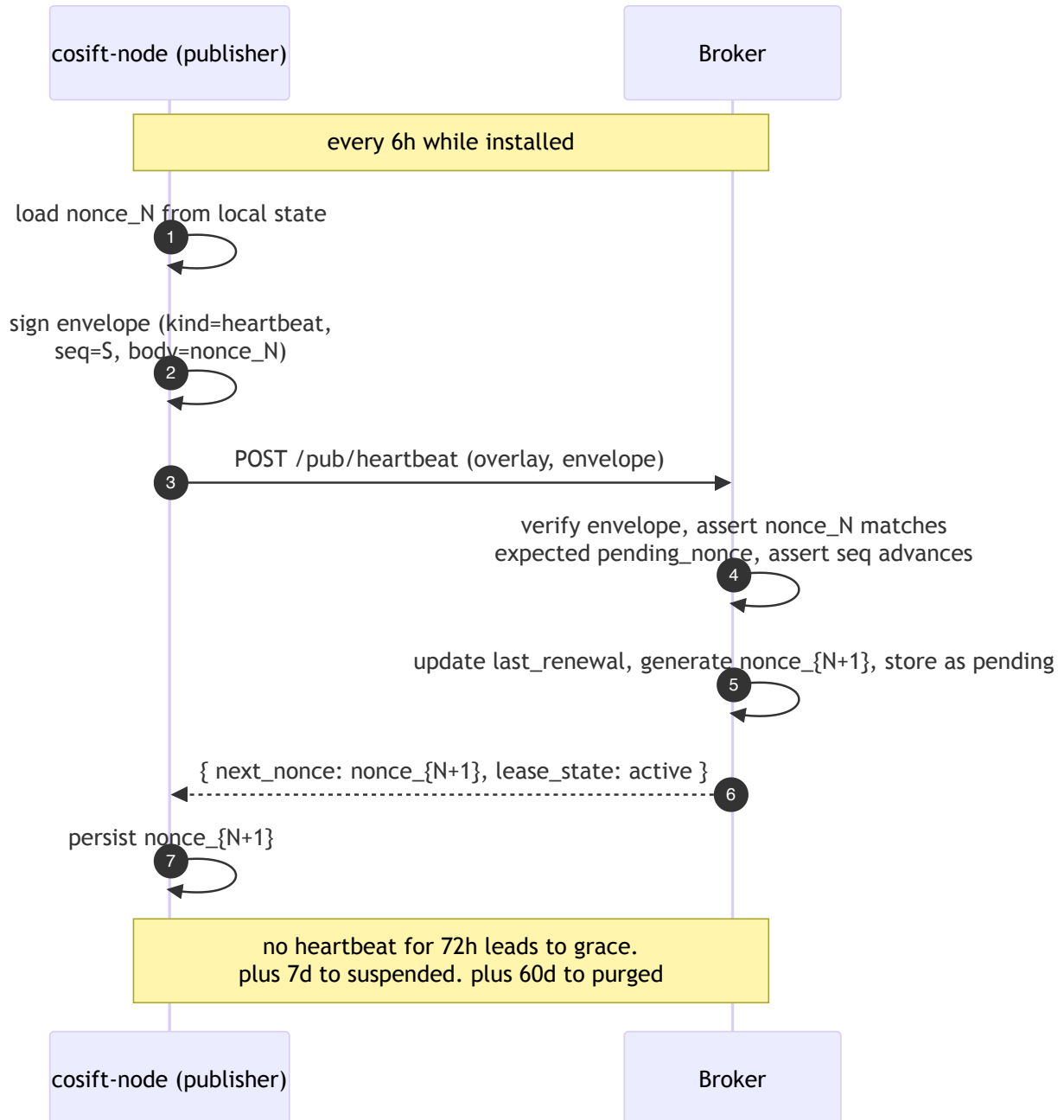
Domain ownership without a central account system. The well-known token is the proof; the broker re-verifies every 30 days.



Conflicting claims: longest-prefix nesting; first verified wins. A second principal claiming the same exact prefix → 409 immediately.

5. Publisher heartbeats and lease state

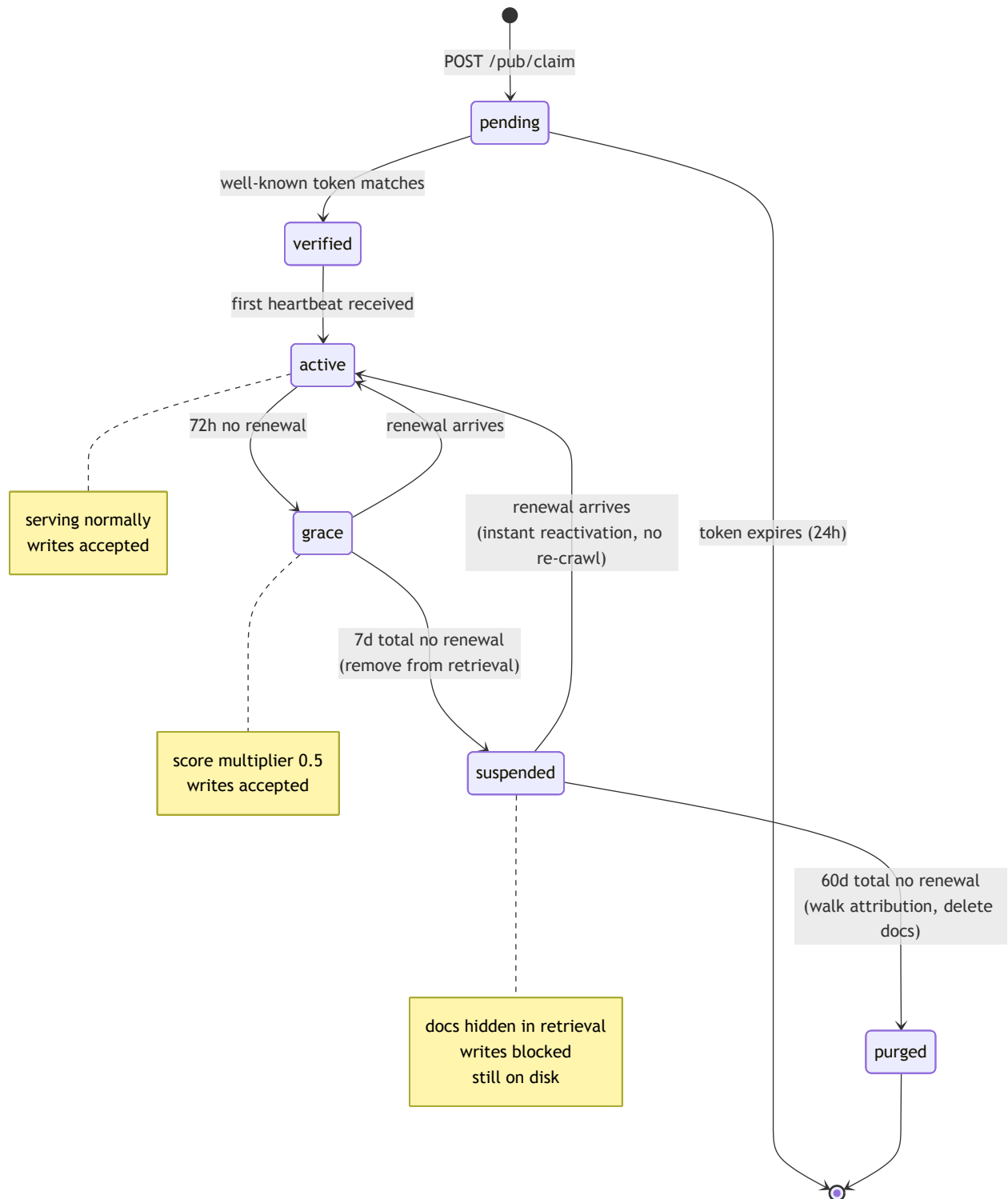
Nonce-chained heartbeats (**D1**) — the response carries the next nonce; replay-proof without inbound delivery. App uninstall = identity destroyed = heartbeats stop = lease lapses.



Sweeper goroutine ticks once a minute; transitions are flag flips on the **'C'** record + an in-memory suspended-set update consumed by retrieval gating (WS6).

6. Lease state machine

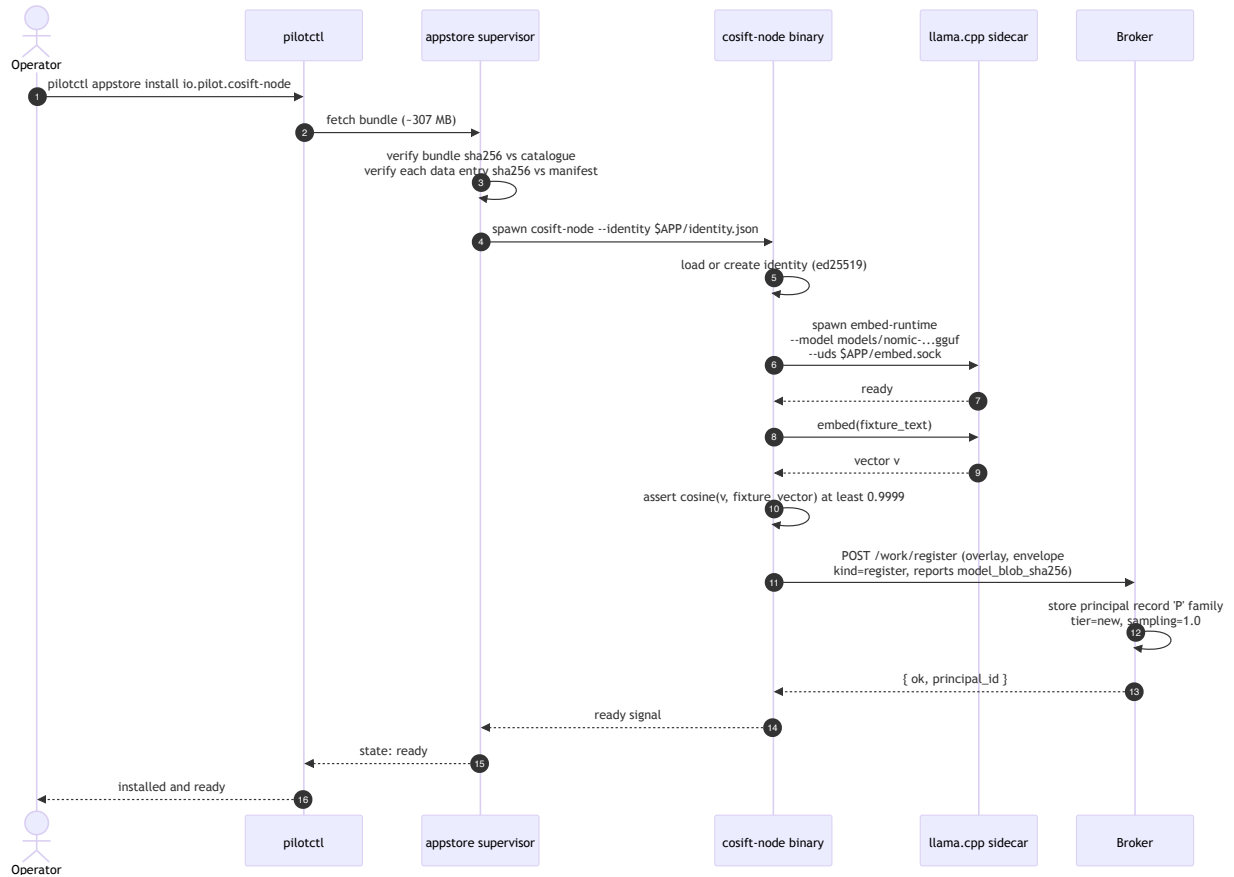
Two state machines live on a collection: the **lease** (presence-driven, this diagram) and the **verification** state (active vs unverified, which gates writes but never serving).



Generous timings: a laptop on vacation doesn't nuke a publisher's docs.

7. Worker install and register

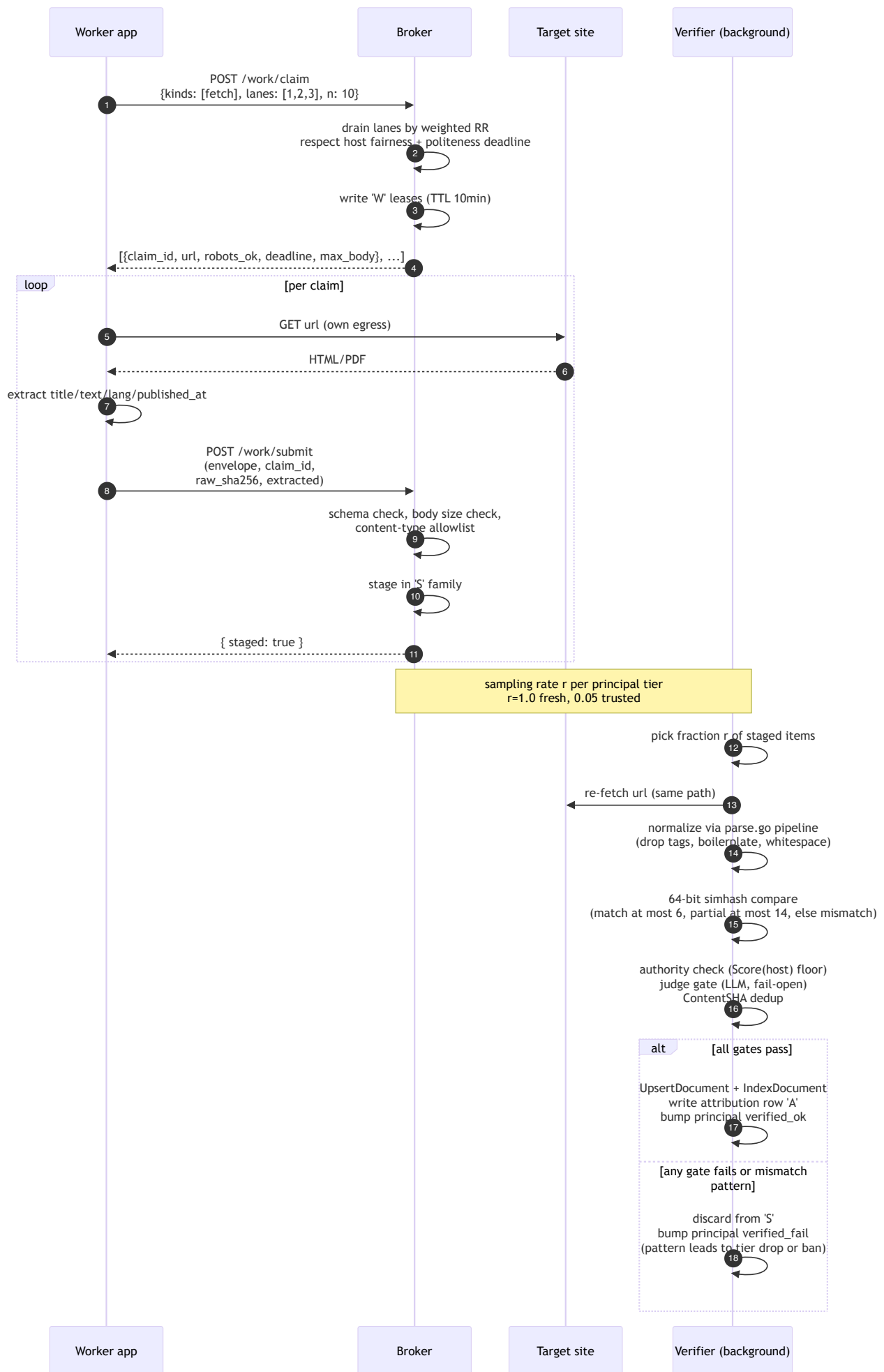
Self-contained — no ollama, no python. The bundle ships the embed runtime and model weights.



If the fixture check fails, the app stays alive for `node.status` but `node.worker_start` returns `runtime_not_ready` — never enters the loop with a tampered model.

8. Fetch contribution lifecycle

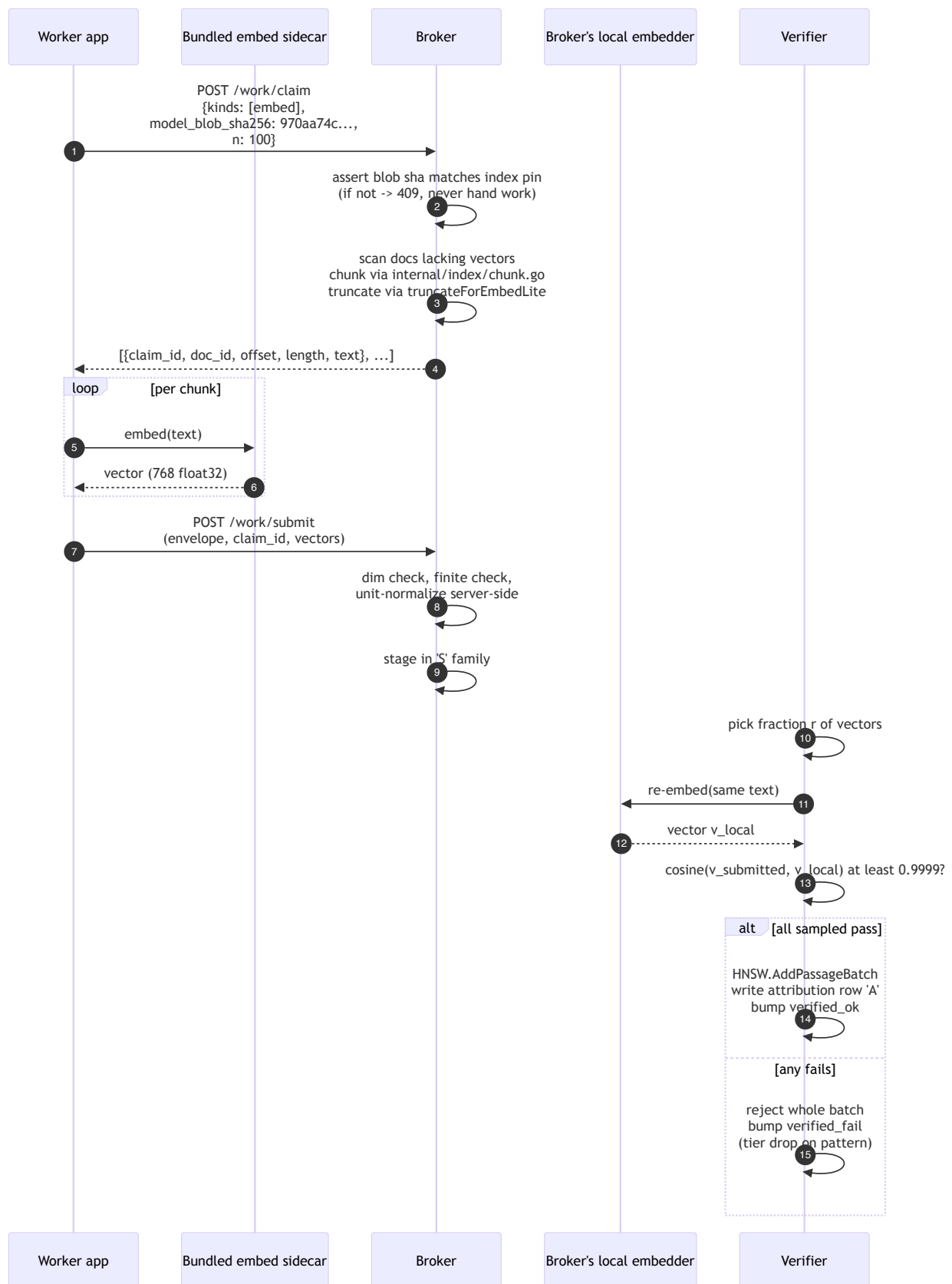
Claim → fetch → submit → quarantine → sampled verify → promote (or reject). Politeness stays server-controlled.



Same churn-aware grading prevents single-doc slashing — a contributor goes down only on a *pattern* of mismatches.

9. Embed contribution lifecycle

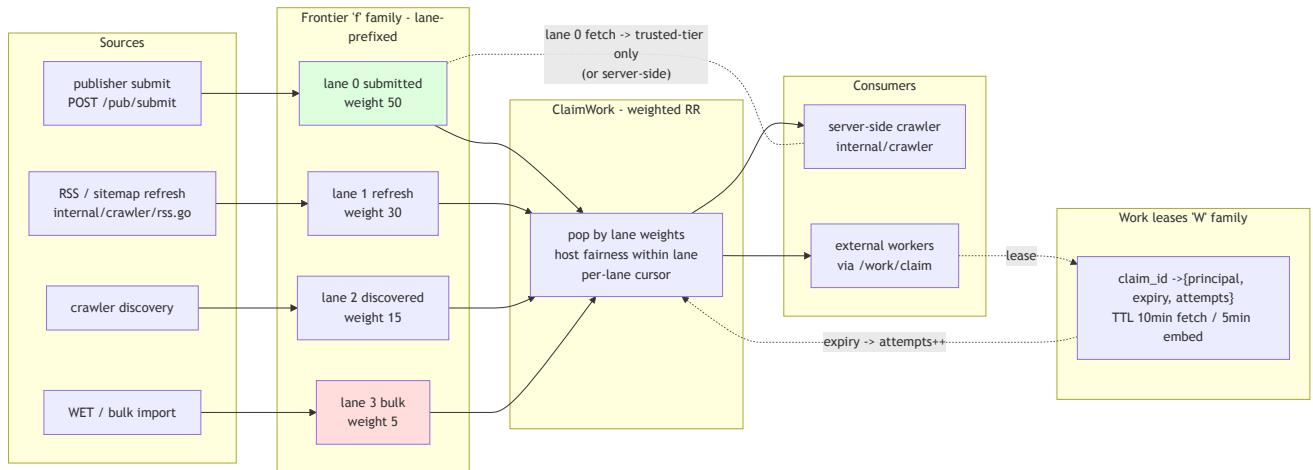
Verification is sharper here than for content — same model + same blob = deterministic match, cosine ≥ 0.9999 .



Server defines the text (**I6**) — contributors never get to choose the input, which collapses the verification problem into "did you compute the right vector for this text."

10. Frontier lanes and work distribution

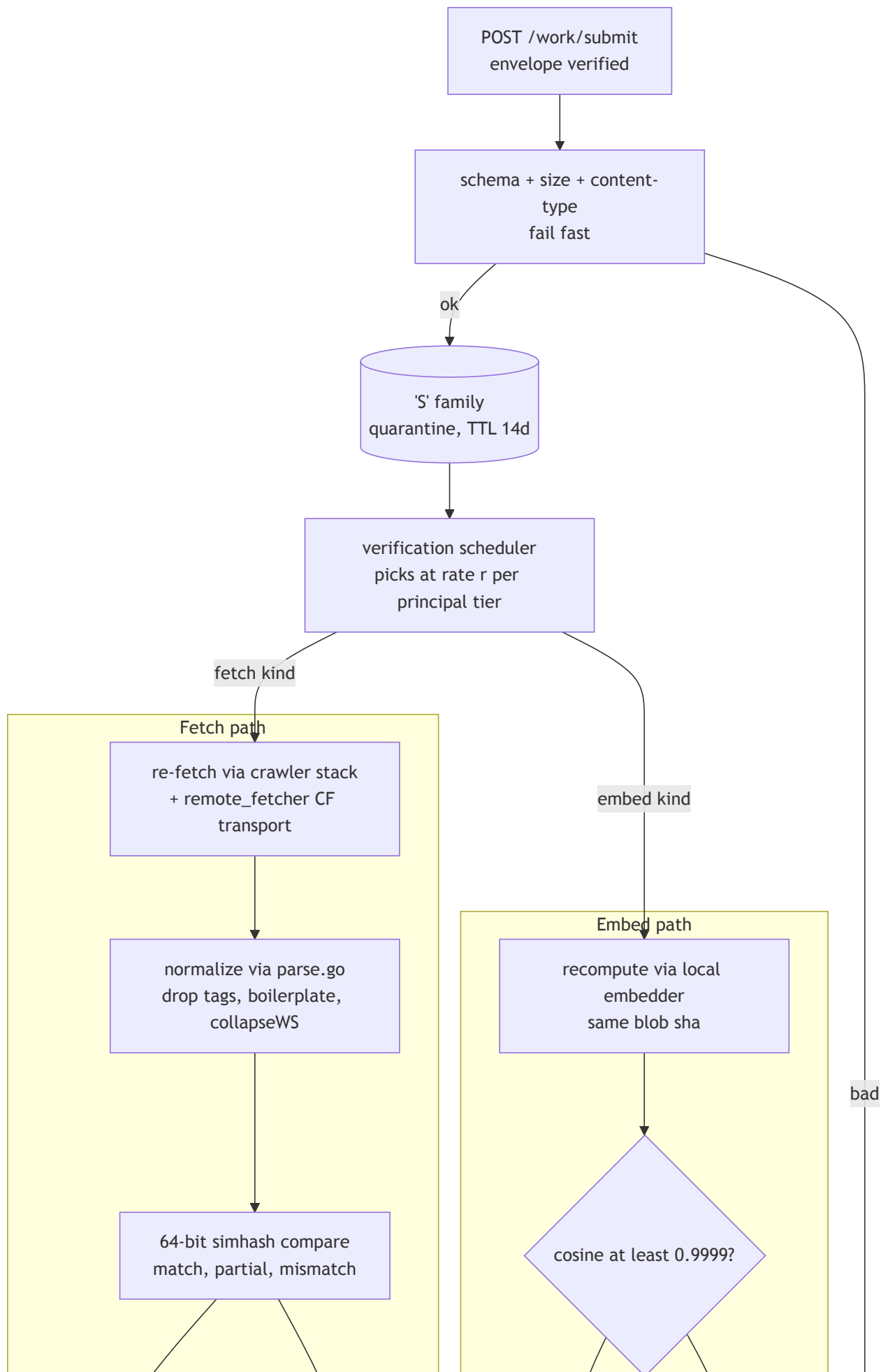
The frontier is a single keyspace with a lane byte; weighted drain prevents bulk from starving submissions.

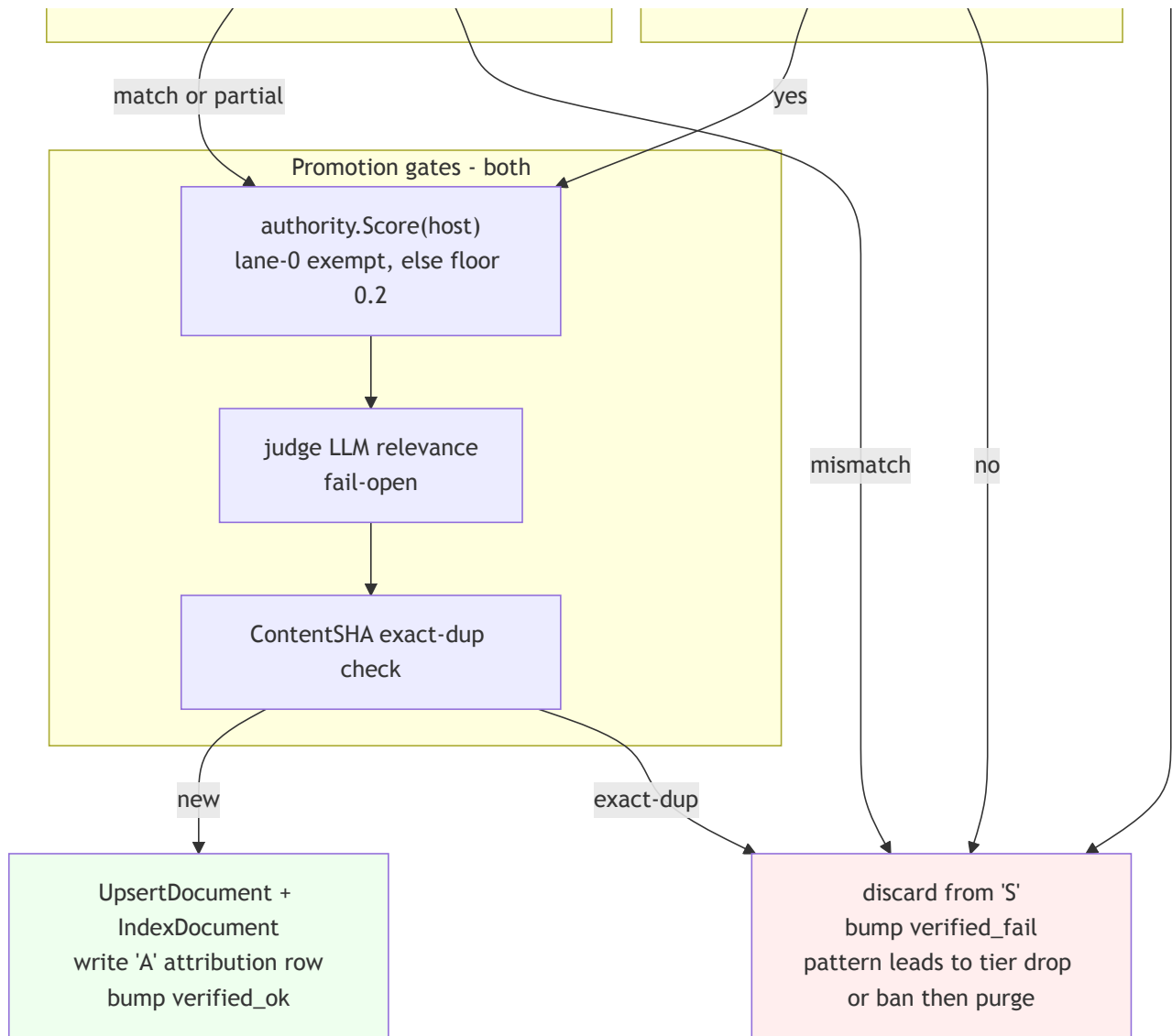


If lane 0 is empty its share redistributes to other lanes; deterministic weighted RR keeps statistical fairness across many claims.

11. Quarantine, verify, promote pipeline

The atomic gate: nothing reaches the main index without passing every check for its kind.

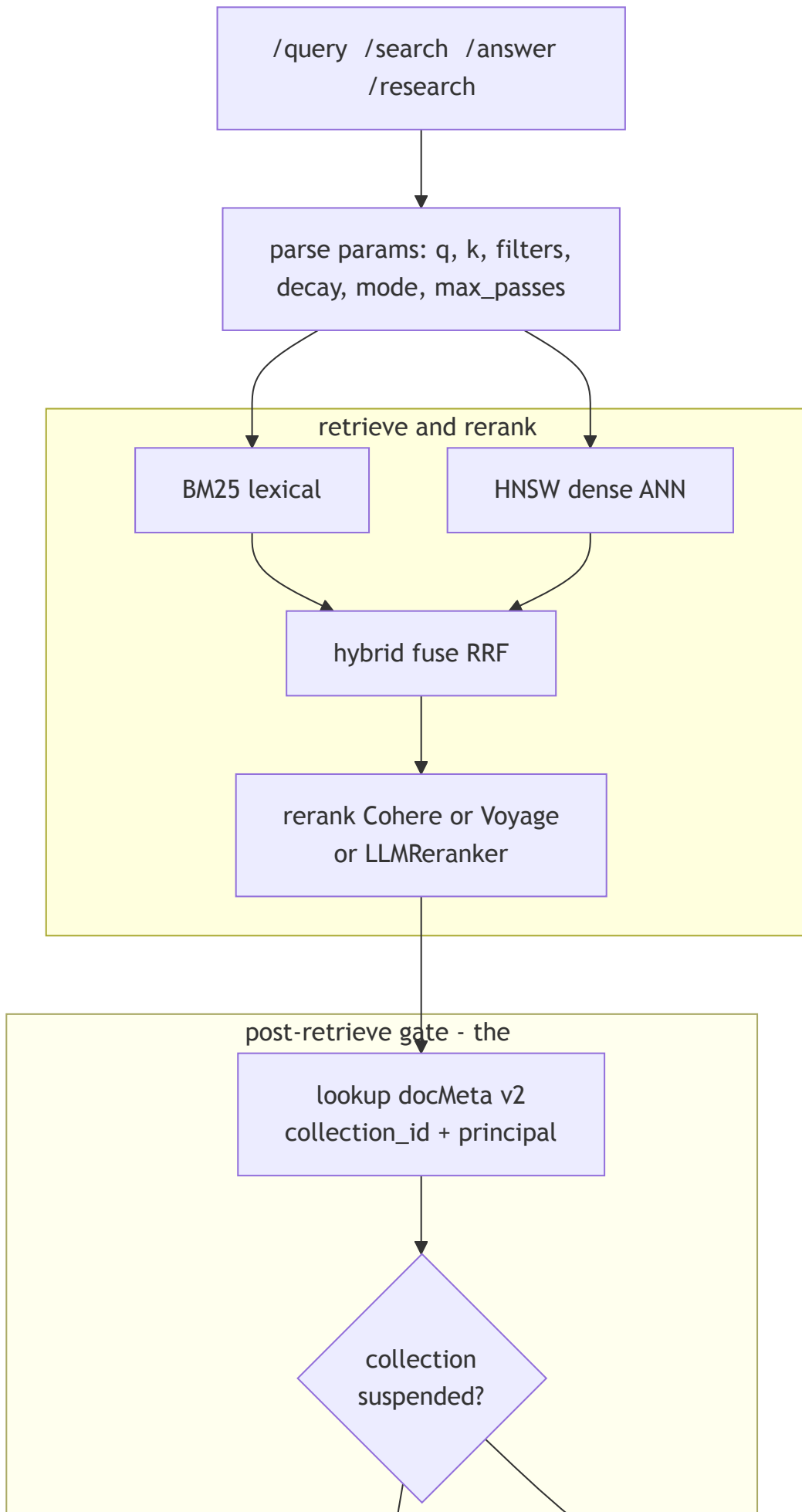


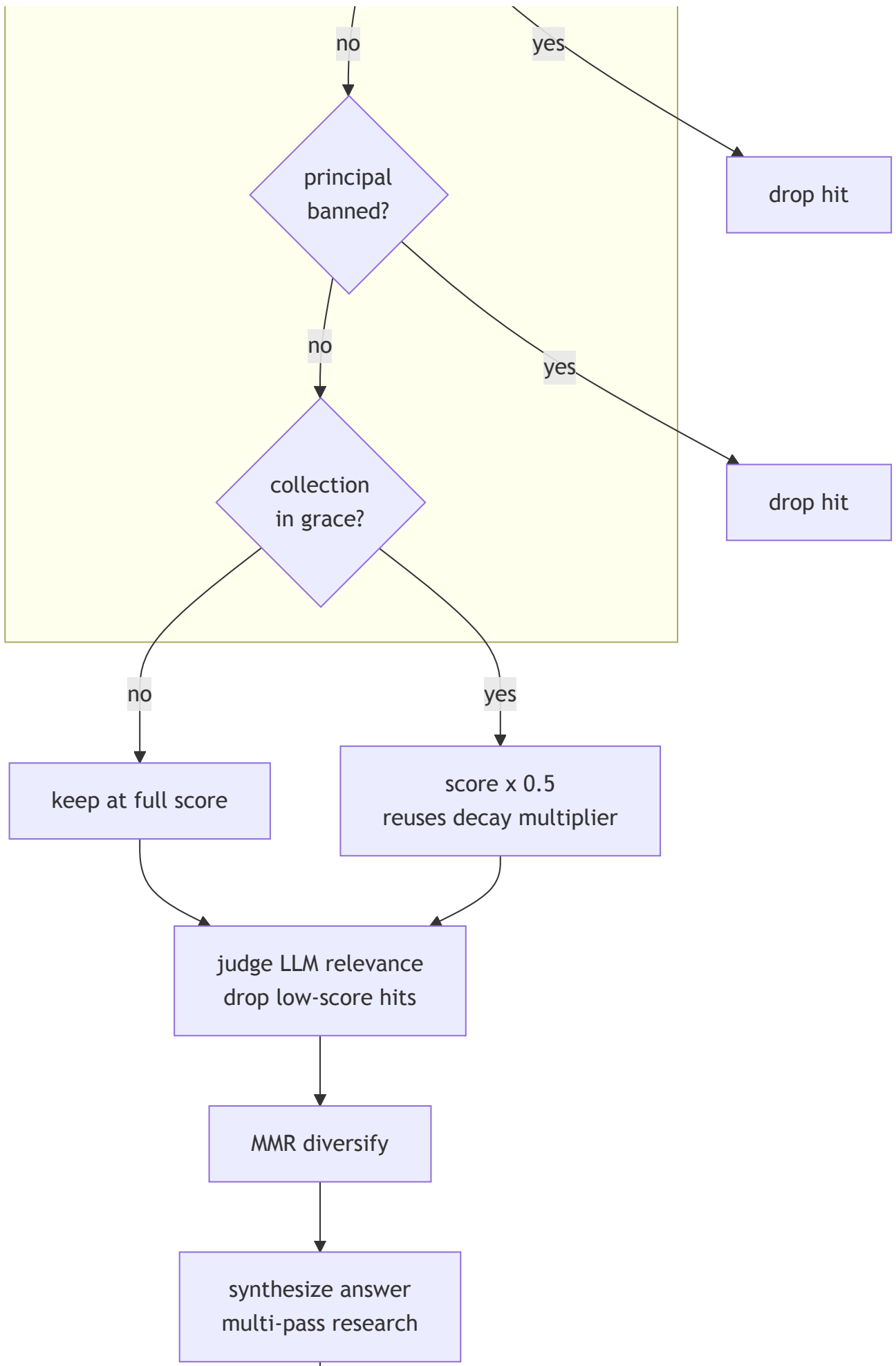


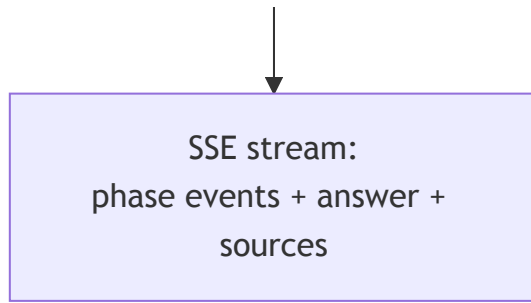
`partial` for content is neutral (page churn — ads, timestamps); only a *pattern* of mismatches drops a tier.

12. Retrieval path with gating

One choke point covers `/search` , `/query` , `/answer` , `/research` : the post- `retrieve()` filter loop. Suspended collections + banned principals filtered there cheaply via the docMeta v2 packed blob.



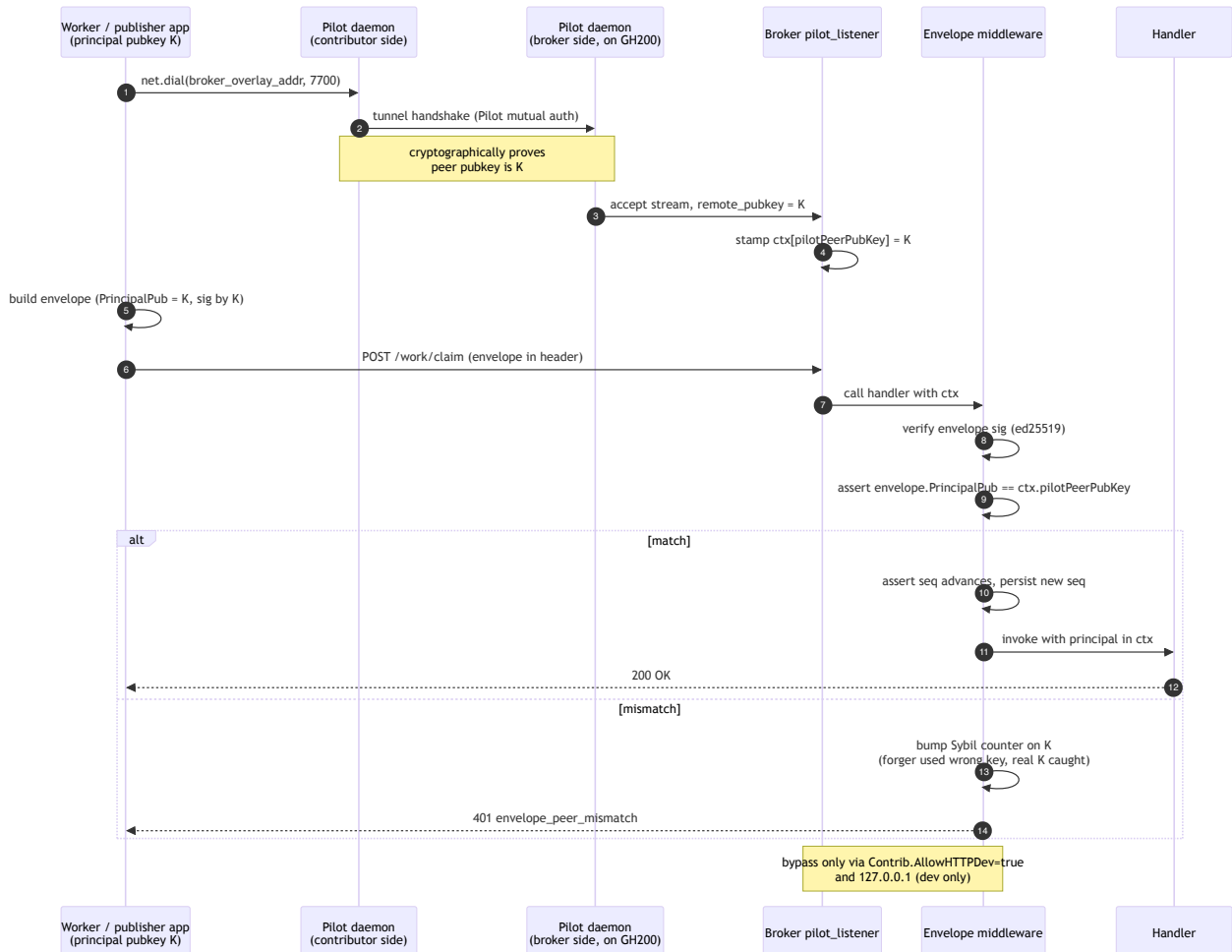




Same machinery for every retrieval endpoint — invariant **I8** enforced by sharing one function, not four.

13. Transport binding (I9)

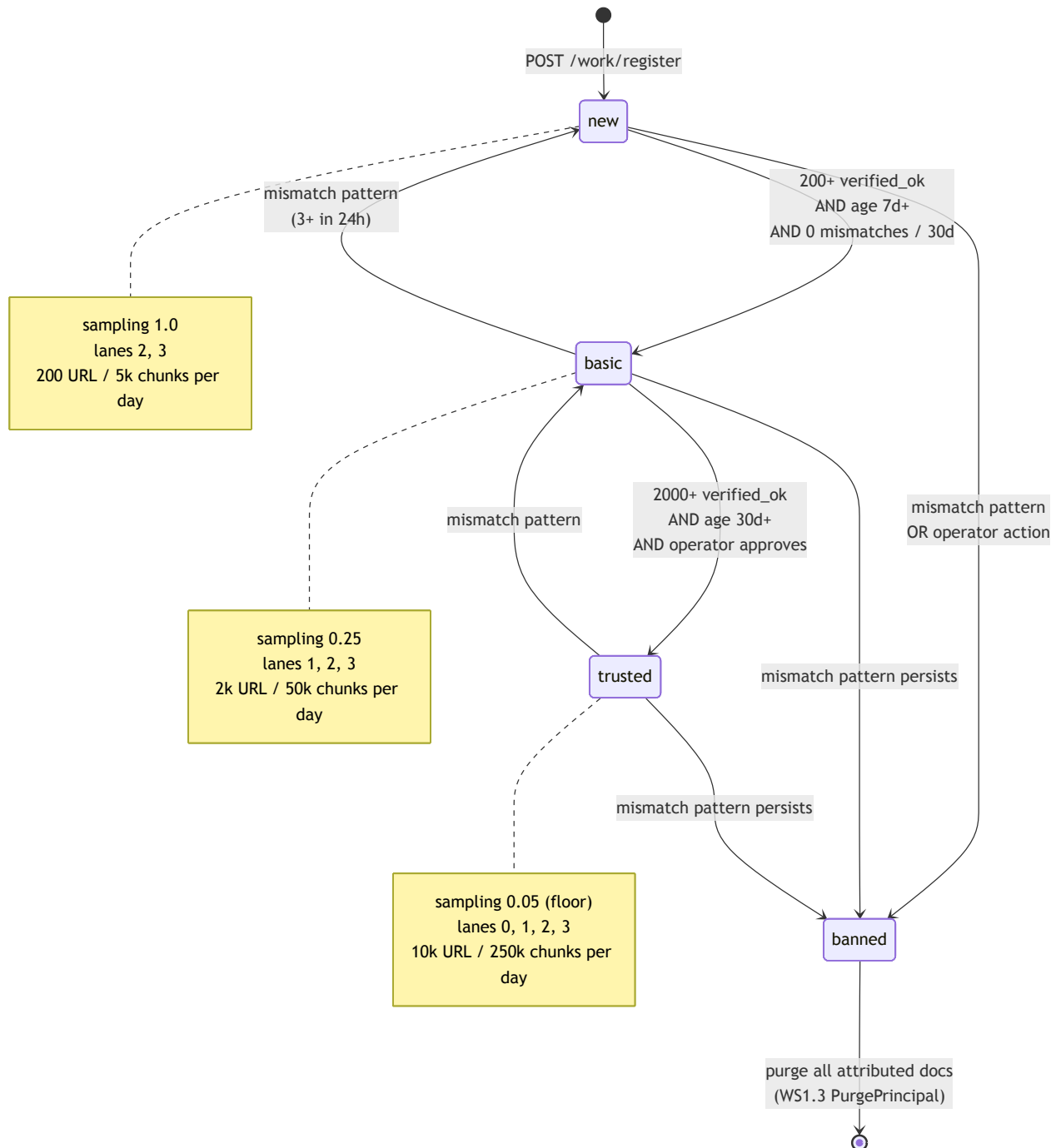
The contribution-path proof: the envelope's signing key must equal the Pilot tunnel's authenticated peer pubkey. Forge either side, broker refuses.



Public HTTPS via Caddy 404s these routes — overlay is the only network-reachable path.

14. Principal reputation

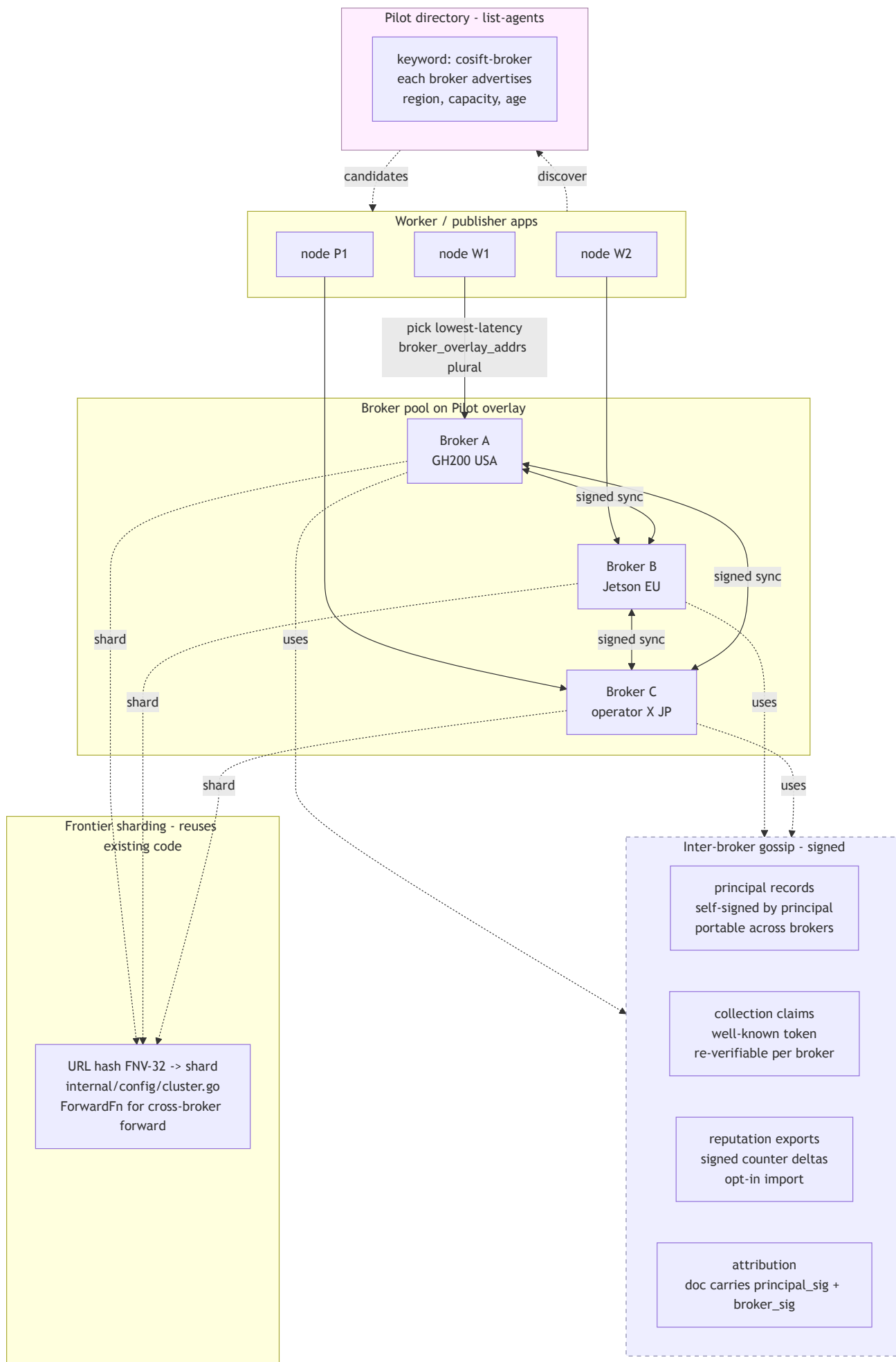
Sampling rate is the lever — fresh keys verified at 100%, earned trust lowers it; mismatch patterns drop you back or ban you.



Expected un-caught bad submissions per burned identity $\approx (1-r)/r \rightarrow 0 / 3 / 19$. Purge (**I4**) reverses survivors.

15. Multi-broker federation (future)

What v2 looks like if **D10** says go. Phase 1-5 keeps the runway clear (WS11a) but ships single-broker.



The portability invariants from WS11a are what make this additive instead of a rewrite: principals, collections, attribution, and frontier sharding all already work cross-broker; v2 just adds discovery + gossip + import logic.

Reading order if you only have ten minutes

1. **Diagram 1** — see the three planes and where Pilot sits.
2. **Diagram 2** — see the developer-to-AI-agent loop end to end.
3. **Diagram 13** — see the I9 transport binding (the security spine).
4. **Diagram 11** — see the verification pipeline (the trust spine).
5. **Diagram 6** — see the lease lifecycle (the adoption flywheel).